

Homework Assignment #2.

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Read Ch 1 Math Preliminaries, pp. 13–19.

Exercise # 1: (10 marks) Find the inverse of the following 4×4 matrix, where the 3×3 matrix is a rotation.

$$\begin{bmatrix} Q & d \\ 0 & 1 \end{bmatrix}$$

Solution. We are looking for a matrix T such that

$$T \cdot \begin{bmatrix} Q & d \\ 0 & 1 \end{bmatrix} = I = \begin{bmatrix} Q & d \\ 0 & 1 \end{bmatrix} T.$$

Since Q is a rotation matrix, $QQ^T = 1$. Also, the 1×1 block matrix has inverse $[1]$. Therefore, T must have the form

$$T = \begin{bmatrix} Q^T & x \\ y^T & 1 \end{bmatrix}$$

for some vectors x, y . We compute:

$$T \cdot \begin{bmatrix} Q & d \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} Q^T & x \\ y^T & 1 \end{bmatrix} \cdot \begin{bmatrix} Q & d \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} I & Q^T d + x \\ y^T Q & 1 \end{bmatrix}.$$

Then, we require $y^T Q = Q^T d + x = 0$. This is the case for an arbitrary rotation matrix Q when $y = 0$ and $x = -Q^T d$. Then, we have:

$$T = \begin{bmatrix} Q^T & -Q^T d \\ 0 & 1 \end{bmatrix}.$$

We check left multiplication:

$$\begin{bmatrix} Q & d \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} Q^T & -Q^T d \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} I & -QQ^T d + d \\ 0 & 1 \end{bmatrix} = I,$$

as desired. Therefore,

$$\begin{bmatrix} Q & d \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} Q^T & -Q^T d \\ 0 & 1 \end{bmatrix}.$$

Exercise # 2: (20 marks)

(i) A vector product function is specified as

$$f(x) = (-2\mathbf{i}_0 + 3\mathbf{j}_0 - 5\mathbf{k}_0) \times x.$$

What is the matrix representation of f in $C_0 = [\mathbf{i}_0 \ \mathbf{j}_0 \ \mathbf{k}_0]$? What is the matrix representation of f in

$$C_1 = [\mathbf{i}_1 \ \mathbf{j}_1 \ \mathbf{k}_1] = [-\mathbf{i}_0 \ -\mathbf{j}_0 \ \mathbf{k}_0]?$$

(ii) What is the coordinate representation 0Q in C_0 of a rotation

$$y = Q(\mathbf{k}_0, \frac{\pi}{2}) x$$

about axis $\mathbf{k}_0$ of angle $\pi/2$?

(iii) What is the coordinate representation 0R in C_0 of a rotation

$$y = R(\mathbf{i}_0, \frac{\pi}{6}) x$$

about axis $\mathbf{i}_0$ of angle $\pi/6$?

(iv) What is the coordinate representation of the composition of the two rotations

$$y = R(\mathbf{i}_0, \frac{\pi}{6}) Q(\mathbf{k}_0, \frac{\pi}{2}) x$$

in C_0 ?

(v) Let C_1 be the rotated frame $C_1 = Q(\mathbf{k}_0, \frac{\pi}{2}) C_0$. What is the coordinate representation of the composition of the two rotations

$$y = R(\mathbf{i}_0, \frac{\pi}{6}) Q(\mathbf{k}_0, \frac{\pi}{2}) x$$

in C_1 ?

Solution.

Solution.

(i) The vector defining the cross product is

$${}^0s = \begin{bmatrix} -2 \\ 3 \\ -5 \end{bmatrix}.$$

The linear map $f(x) = s \times x$ is represented in C_0 by the skew-symmetric matrix

$${}^0f = ({}^0s)^\times = \begin{bmatrix} 0 & 5 & 3 \\ -5 & 0 & 2 \\ -3 & -2 & 0 \end{bmatrix}.$$

The change-of-coordinates matrix between C_0 and C_1 is

$${}^0C_1 = \text{diag}(-1, -1, 1), \quad {}^1C_0 = ({}^0C_1)^T = {}^0C_1.$$

The representation of the same linear map in C_1 is therefore

$${}^1f = {}^1C_0 {}^0f {}^0C_1 = \begin{bmatrix} 0 & 5 & -3 \\ -5 & 0 & -2 \\ 3 & 2 & 0 \end{bmatrix}.$$

(ii) A rotation of angle $\pi/2$ about $\mathbf{k}_0$ has coordinate representation

$${}^0Q = \begin{bmatrix} \cos \frac{\pi}{2} & -\sin \frac{\pi}{2} & 0 \\ \sin \frac{\pi}{2} & \cos \frac{\pi}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(iii) A rotation of angle $\pi/6$ about $\mathbf{i}_0$ has coordinate representation

$${}^0R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \frac{\pi}{6} & -\sin \frac{\pi}{6} \\ 0 & \sin \frac{\pi}{6} & \cos \frac{\pi}{6} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}.$$

(iv) The coordinate representation of the composition RQ in C_0 is

$${}^0(RQ) = {}^0R {}^0Q = \begin{bmatrix} 0 & -1 & 0 \\ \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{bmatrix}.$$

(v) Since $C_1 = Q C_0$, we have ${}^0C_1 = {}^0Q$ and ${}^1C_0 = ({}^0Q)^T$. Thus the coordinate representation of the same rotation in C_1 is

$${}^1(RQ) = {}^1C_0 {}^0(RQ) {}^0C_1 = \begin{bmatrix} 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 1 & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}.$$

Exercise # 3 (10 marks)

Find the velocity of a point $\tilde{y}$ with coordinates

$$y(t) = d(t) + Q(t) x(t),$$

where $Q(t)$ is a rotation matrix, as a function of $\dot{d}$, $\dot{x}$ and $\dot{Q} = \omega \times Q$. Provide a physical interpretation for each of the terms. Find the acceleration of a point with coordinates

$$y(t) = d(t) + Q(t) x(t),$$

where $Q(t)$ is a rotation matrix.

Solution.

We compute:

$$\dot{y}(t) = \dot{d}(t) + \dot{Q}(t)x(t) + Q(t)\dot{x}(t) = \dot{d}(t) + \omega \times Q(t)x(t) + Q(t)\dot{x}(t).$$

If $y(t)$ is in frame C_1 , then let $C_1 = Q C_0$. That is, $Q = {}^1 C_0$.

- $\dot{x}$ is the velocity of the point with respect to the co-ordinate origin d in frame C_0
- ω is the angular velocity of frame C_0 with respect to frame C_1
- $\dot{d}$ is the velocity of frame C_0 w.r.t. the frame C_1 .

We compute the acceleration:

$$\begin{aligned} \ddot{y} &= \ddot{d} + \dot{\omega} \times Qx + \omega \times \omega \times Qx + \omega \times Q\dot{x} + \omega \times Q\dot{x} + Q\ddot{x} \\ &= \ddot{d} + \dot{\omega} \times Qx + \omega \times \omega \times Qx + 2\omega \times Q\dot{x} + Q\ddot{x}. \end{aligned}$$

where we used the fact that $\dot{Q} = \omega \times Q$.

Only graduate students have to hand in the following two problems:

Exercise # 4: (30 marks)

Read the ch1 Quaternions2024.pdf chapter from the Reading Material Canvas Module.

Write a Simulink model to integrate

$$\dot{Q} = \omega \times Q, \quad \dot{\omega} = \tau, \quad \omega(0) = 0, \quad Q(0) = I,$$

driven by a specified torque τ . Animate the motion (block provided in Simulink tutorial, can use a frame or other) whose coordinate transformation is given by Q . This corresponds to your animating a rigid body having an inertia matrix equal to the identity. Does the matrix $Q(t)$ remain a rotation matrix for all t as you integrate it? If not, how would you fix it?

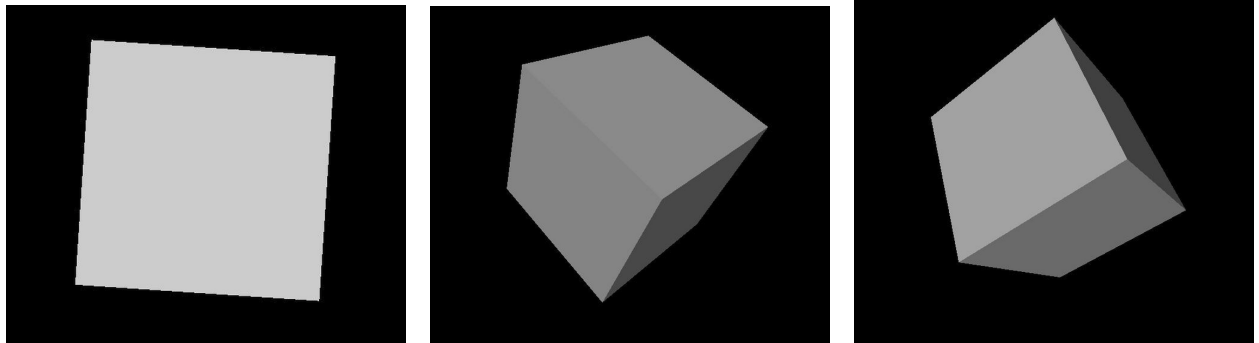
Provide $Q(t)$ and the animation (or just last image) for the following cases:

- (a) $\tau = [0; 0; 1]$, $t = 5$ s;
- (b) $\tau = [0; -1; 1]$, $t = 8$ s;
- (c) $\tau = [1; 1; -1]$, $t = 4$ s.

Write a Simulink block to integrate the quaternion, with the same τ and ω as in Exercise # 3. Generate the rotation matrix, animate the motion. How about $q(t)$, does it remain a “unit” quaternion? Propose and implement a method to fix this. Discuss.

Solution.

$Q(t)$ and last image for integrated rotation matrix:



(a) $t = 5$ s, $\tau = [0 \ 0 \ 1]$

(b) $t = 8$ s, $\tau = [0 \ -1 \ 1]$

(c) $t = 4$ s, $\tau = [1 \ 1 \ -1]$

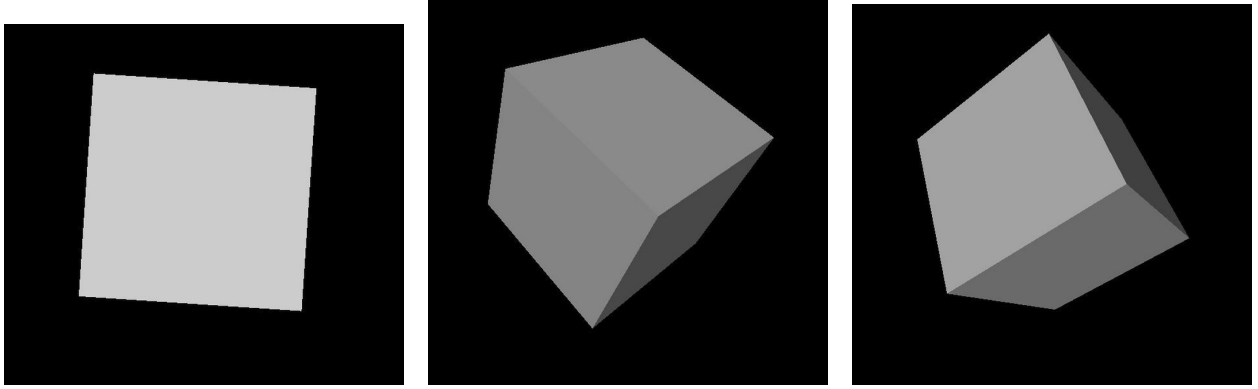
Figure 1: Rigid-body orientation using rotation-matrix integration (Q -integration).

- (a) $\tau = [0 \ 0 \ 1]$, $t = 5$ s.

$$Q(5) = \begin{bmatrix} 0.9978 & 0.0663 & 0 \\ -0.0663 & 0.9978 & 0 \\ 0 & 0 & 1.0000 \end{bmatrix}$$

$$q(5) = 0.9994 + 0i + 0j - 0.0332k$$

$$R(q(5)) \approx \begin{bmatrix} 0.9978 & 0.0664 & 0 \\ -0.0664 & 0.9978 & 0 \\ 0 & 0 & 1.0000 \end{bmatrix}$$



(a) $t = 5 \text{ s}$, $\tau = [0 \ 0 \ 1]$

(b) $t = 8 \text{ s}$, $\tau = [0 \ -1 \ 1]$

(c) $t = 4 \text{ s}$, $\tau = [1 \ 1 \ -1]$

Figure 2: Rigid-body orientation using quaternion integration with normalization.

(b) $\tau = [0 \ -1 \ 1]$, $t = 8 \text{ s}$.

$$Q(8) = \begin{bmatrix} 0.2902 & -0.6760 & -0.6760 \\ 0.6760 & 0.6451 & -0.3549 \\ 0.6760 & -0.3549 & 0.6451 \end{bmatrix}$$

$$q(8) = -0.8043 + 0i + 0.4202j - 0.4202k$$

$$R(q(8)) \approx \begin{bmatrix} 0.2938 & -0.6759 & -0.6759 \\ 0.6759 & 0.6469 & -0.3531 \\ 0.6759 & -0.3531 & 0.6469 \end{bmatrix}$$

(c) $\tau = [1 \ 1 \ -1]$, $t = 4 \text{ s}$.

$$Q(4) = \begin{bmatrix} 0.5180 & 0.7957 & 0.3137 \\ -0.3137 & 0.5180 & -0.7957 \\ -0.7957 & 0.3137 & 0.5180 \end{bmatrix}$$

$$q(4) = 0.7991 + 0.3471i + 0.3471j - 0.3471k$$

$$R(q(4)) \approx \begin{bmatrix} 0.5181 & 0.7957 & 0.3138 \\ -0.3138 & 0.5181 & -0.7957 \\ -0.7957 & 0.3138 & 0.5181 \end{bmatrix}$$

The matrix $Q(t)$ does not remain a rotation matrix for all t as you integrate it because of numerical error in the integration. We can confirm this pretty easily: indeed, a rotation matrix has the property that $Q \cdot Q^T = I$. But, for $Q(5)$ in part a), we have

$$Q(5)(Q(5))^T = \begin{bmatrix} 1.00000053 & 0 & 0 \\ 0 & 1.00000053 & 0 \\ 0 & 0 & 1 \end{bmatrix} \neq I.$$

Therefore, $Q(5)$ is not a perfect rotation matrix. This could be rectified by making the singular values of the matrix 1. If we take the SVD composition of Q , we will get $Q = U\Sigma V^T$. U and V^T are orthonormal matrices, and Σ will be a diagonal matrix with singular values being close to 1,

accounting for the error. If we reset the singular values all to 1, then $Q_{fixed} = UV^T$ will be an orthogonal matrix.

$q(t)$ also does not remain a "unit" quaternion, because of numerical integration errors. Indeed, we can compute $q(5)q(5)^* = 0.99990260 \neq 1$. In order to rectify this, we can normalize the quaternion after each integration.

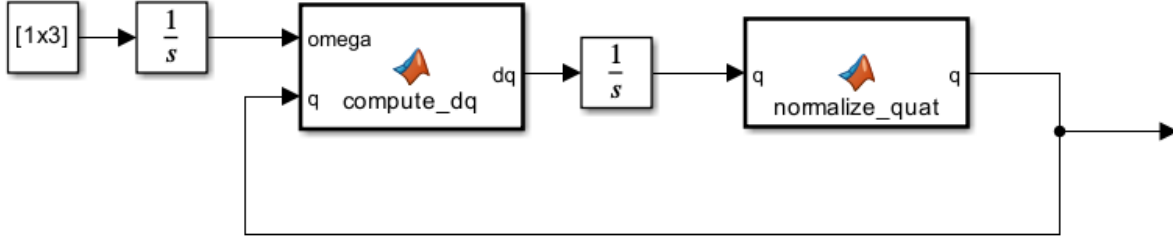


Figure 3: Simulink Block to integrate and normalize quaternion, so that quaternion remains a "unit" rotation quaternion

The code block in "normalize_quat" simply computes $q = q/\|q\|$.

Exercise # 5: (20 marks)

- (a) Show that for any $x \in \mathbb{R}^4$,

$$x^T e^{\theta(\hat{s}^\circ)} x = \cos \theta \|x\|^2,$$

where $e^{\theta(\hat{s}^\circ)}$ is as given in equation (34), p. 6 of the quaternion handout.

- (b) Consider the function $f : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ defined by the quaternion product

$$f(x) = (d + a\mathbf{i})x(d + a\mathbf{i})^{-1}$$

(or in vector form $(d + a\mathbf{i}) \circ (dx + v_x) \circ (d + a\mathbf{i})^{-1}$, similarly to eq (45) p. 9 of the notes on Quaternions, but with x and $f(x)$ in $\mathbb{R}^4$). Also define

$$f_\ell(x) = (d + a\mathbf{i})x, \quad f_r(x) = x(d + a\mathbf{i})^{-1}, \quad f_\ell, f_r : \mathbb{R}^4 \rightarrow \mathbb{R}^4.$$

Clearly f_ℓ , f_r , and f are linear. Find their matrix representations. What do each of these functions represent, geometrically?

Solution.

1. We are given that

$$e^{\theta(\hat{s}^\circ)} = \cos \theta I + \sin \theta \begin{bmatrix} 0 & -\hat{s}^T \\ \hat{s} & \hat{s} \times \end{bmatrix}.$$

Write $x = [x_0 \ \hat{x}]^T$ where $\hat{x} = [x_1 \ x_2 \ x_3]^T$. Then, we compute:

$$\begin{aligned}
x^T e^{\theta(\hat{s} \circ)} x &= x^T (\cos \theta I + \sin \theta \begin{bmatrix} 0 & -\hat{s}^T \\ \hat{s} & \hat{s} \times \end{bmatrix}) x \\
&= \cos \theta x^T x + \sin \theta x^T \begin{bmatrix} 0 & -\hat{s}^T \\ \hat{s} & \hat{s} \times \end{bmatrix} x \\
&= \cos \theta \|x\|^2 + \sin \theta x^T \begin{bmatrix} -\hat{s}^T \hat{x} \\ x_0 \hat{s} + \hat{s} \times \hat{x} \end{bmatrix} \\
&= \cos \theta \|x\|^2 + \sin \theta (-x_0 \hat{s}^T \hat{x} + \hat{x}^T (x_0 \hat{s}) + \hat{x}^T \hat{s} \times \hat{x}).
\end{aligned}$$

However, $\hat{x}^T \hat{s} \times \hat{x} = 0$ since $\hat{x}^T$ and $\hat{s} \times \hat{x}$ are perpendicular so their dot product is zero. Also, $-x_0 \hat{s}^T \hat{x} + \hat{x}^T (x_0 \hat{s}) = 0$. Thus, the $\sin \theta$ term is zero so we get $x^T e^{\theta(\hat{s} \circ)} x = \cos \theta \|x\|^2$.

2. First, we interpret a vector $x \in \mathbb{R}^4$ as $[x_0 \ x_1 \ x_2 \ x_3] = x_0 + x_1 i + x_2 j + x_3 k$. Also, we recall that $(d + ai)^{-1} = \frac{1}{d^2 + a^2} [d - ai]$. Then,

$$\begin{aligned}
f_l(x) &= (d + ai)(x_0 + x_1 i + x_2 j + x_3 k) \\
&= (dx_0 - ax_1) + (dx_1 + ax_0)i + (dx_2 - ax_3)j + (dx_3 + ax_2)k \\
&= \begin{bmatrix} d & -a & 0 & 0 \\ a & d & 0 & 0 \\ 0 & 0 & d & -a \\ 0 & 0 & a & d \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} \\
&= \sqrt{a^2 + d^2} \begin{bmatrix} \frac{d}{\sqrt{a^2 + d^2}} & -\frac{a}{\sqrt{a^2 + d^2}} & 0 & 0 \\ \frac{a}{\sqrt{a^2 + d^2}} & \frac{d}{\sqrt{a^2 + d^2}} & 0 & 0 \\ 0 & 0 & \frac{d}{\sqrt{a^2 + d^2}} & -\frac{a}{\sqrt{a^2 + d^2}} \\ 0 & 0 & \frac{a}{\sqrt{a^2 + d^2}} & \frac{d}{\sqrt{a^2 + d^2}} \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}
\end{aligned}$$

In this form, we can see that this matrix performs two rotations, in the i_0, i_1 and i_2, i_3 planes respectively, both "counter-clockwise" by angle $\theta = \cos^{-1}(d/\sqrt{a^2 + d^2})$, and scales the rotated vector by $\sqrt{a^2 + d^2}$.

Similarly,

$$\begin{aligned}
f_r(x) &= (x_0 + x_1 i + x_2 j + x_3 k) \cdot \frac{1}{d^2 + a^2} (d - ai) \\
&= \frac{1}{d^2 + a^2} ((x_0 d + ax_1) + (dx_1 - ax_0)i + (dx_2 - ax_3)j + (dx_3 + ax_2)k) \\
&= \frac{1}{d^2 + a^2} \begin{bmatrix} d & a & 0 & 0 \\ -a & d & 0 & 0 \\ 0 & 0 & d & -a \\ 0 & 0 & a & d \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} \\
&= \frac{1}{\sqrt{d^2 + a^2}} \begin{bmatrix} \frac{d}{\sqrt{a^2 + d^2}} & \frac{a}{\sqrt{a^2 + d^2}} & 0 & 0 \\ -\frac{a}{\sqrt{a^2 + d^2}} & \frac{d}{\sqrt{a^2 + d^2}} & 0 & 0 \\ 0 & 0 & \frac{d}{\sqrt{a^2 + d^2}} & -\frac{a}{\sqrt{a^2 + d^2}} \\ 0 & 0 & \frac{a}{\sqrt{a^2 + d^2}} & \frac{d}{\sqrt{a^2 + d^2}} \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}.
\end{aligned}$$

We likewise recognize this as a linear map that performs a clockwise rotation in the plane i_0, i_1 , and a counter-clockwise rotation in the plane i_2, i_3 , both by angle $\theta = \cos^{-1}(d/\sqrt{a^2 + d^2})$. Then, we compute $f = f_l \cdot f_r$.

$$f = \frac{1}{a^2 + d^2} \begin{bmatrix} a^2 + d^2 & 0 & 0 & 0 \\ 0 & a^2 + d^2 & 0 & 0 \\ 0 & 0 & d^2 - a^2 & -2ad \\ 0 & 0 & 2ad & d^2 - a^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{d^2 - a^2}{d^2 + a^2} & \frac{-2ad}{d^2 + a^2} \\ 0 & 0 & \frac{2ad}{d^2 + a^2} & \frac{d^2 - a^2}{d^2 + a^2} \end{bmatrix}.$$

Geometric interpretation: We recognize the trig identities: $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$, and $\sin(2\theta) = 2 \sin \theta \cos \theta$. Then, if $\cos \theta = \frac{d}{\sqrt{a^2 + d^2}}$, we have $\cos(2\theta) = \frac{d^2 - a^2}{d^2 + a^2}$ and $\sin(2\theta) = \frac{2ad}{d^2 + a^2}$. Therefore, this matrix performs the identity in the plane defined by i_0, i_1 , and a rotation by $2 \cos^{-1}(d/\sqrt{a^2 + d^2})$ in the counter-clockwise direction.